

12 Functions

12.1 Functions

2. Suppose $A = \{a, b, c, d\}$, $B = \{2, 3, 4, 5, 6\}$ and

$$f = \{(a, 2), (b, 3), (c, 4), (d, 5)\}.$$

State the domain and range of f . Find $f(b)$ and $f(d)$.

The domain of f is A and the range of f is $\{2, 3, 4, 5\}$.
 $f(b) = 3$ and $f(d) = 5$.

4. There are eight different functions $f : \{a, b, c\} \rightarrow \{0, 1\}$. List them all. Diagrams will suffice.

$$\begin{aligned} f_1 &= \{(a, 0), (b, 0), (c, 0)\} \\ f_2 &= \{(a, 0), (b, 0), (c, 1)\} \\ f_3 &= \{(a, 0), (b, 1), (c, 0)\} \\ f_4 &= \{(a, 0), (b, 1), (c, 1)\} \\ f_5 &= \{(a, 1), (b, 0), (c, 0)\} \\ f_6 &= \{(a, 1), (b, 0), (c, 1)\} \\ f_7 &= \{(a, 1), (b, 1), (c, 0)\} \\ f_8 &= \{(a, 1), (b, 1), (c, 1)\} \end{aligned}$$

6. Suppose $f : \mathbb{Z} \rightarrow \mathbb{Z}$ is defined as $f = \{(x, 4x + 5) : x \in \mathbb{Z}\}$. State the domain, codomain and range of f . Find $f(10)$.

The domain and codomain of f are both $\mathbb{Z}$. The range is the set of values $4x + 5$ for $x \in \mathbb{Z}$, that is, the integers that are one more than a multiple of 4:

$$\{4x + 5 : x \in \mathbb{Z}\} = \{\dots, -7, -3, 1, 5, 9, 13, \dots\}.$$

8. Consider the set $f = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x + 3y = 4\}$. Is this a function from $\mathbb{Z}$ to $\mathbb{Z}$? Explain.

No. For f to be a function from $\mathbb{Z}$ to $\mathbb{Z}$, every integer must occur as the first coordinate of some pair in f . But $x = 0$ does not: that would require $3y = 4$, which has no integer solution y . So 0 is not the first coordinate of any element of f , and f is not a function from $\mathbb{Z}$ to $\mathbb{Z}$.

10. Consider the set $f = \{(x^3, x) : x \in \mathbb{R}\}$. Is this a function from $\mathbb{R}$ to $\mathbb{R}$? Explain.

Yes, this is a function $f : \mathbb{R} \rightarrow \mathbb{R}$. Every real number k is the cube of exactly one real number, namely $x = \sqrt[3]{k}$, so each $k \in \mathbb{R}$ occurs exactly once as the first coordinate of an ordered pair in f .

12. Is the set $\theta = \{((x, y), (3y, 2x, x + y)) : x, y \in \mathbb{R}\}$ a function? If so, what is its domain, codomain and range?

Yes. Every first coordinate is a pair $(x, y) \in \mathbb{R}^2$, and each such pair occurs exactly once, so θ is a function $\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ given by $\theta(x, y) = (3y, 2x, x + y)$. Its domain is $\mathbb{R}^2$ and its codomain is $\mathbb{R}^3$.

For the range, a triple $(a, b, c) \in \mathbb{R}^3$ equals $\theta(x, y)$ exactly when $a = 3y$, $b = 2x$ and $c = x + y$, that is, when $y = \frac{a}{3}$, $x = \frac{b}{2}$ and $c = \frac{b}{2} + \frac{a}{3}$. Conversely, given any (a, b, c) with $c = \frac{a}{3} + \frac{b}{2}$, taking $x = \frac{b}{2}$ and $y = \frac{a}{3}$ gives $\theta(x, y) = (a, b, c)$. Therefore the range is

$$\{(a, b, c) \in \mathbb{R}^3 : 2a + 3b = 6c\},$$

a plane through the origin in $\mathbb{R}^3$.

12.2 Injective and Surjective Functions

2. Consider the logarithm function $\ln : (0, \infty) \rightarrow \mathbb{R}$. Decide whether this function is injective and whether it is surjective.

This function is bijective. Observe that, by properties of the exponential function, $e^{\ln x} = x$ for every $x \in (0, \infty)$, and $\ln e^b = b$ for every $b \in \mathbb{R}$.

First we show $\ln$ is injective. Suppose $\ln m = \ln n$ for some $m, n \in (0, \infty)$. Applying the exponential function to both sides gives $e^{\ln m} = e^{\ln n}$, that is, $m = n$. Thus $\ln$ is injective.

Next we show $\ln$ is surjective. Take any $b \in \mathbb{R}$, and let $a = e^b$. Then $a \in (0, \infty)$, since $e^b > 0$ for every real b , and $\ln a = \ln e^b = b$. So every $b \in \mathbb{R}$ equals $\ln a$ for some a in the domain, and $\ln$ is surjective.

Therefore $\ln$ is bijective since it is both injective and surjective on this domain.

4. A function $f : \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$ is defined as $f(n) = (2n, n + 3)$. Verify whether this function is injective and whether it is surjective.

This function is injective but not surjective.

First we show f is injective. Suppose $f(m) = f(n)$ for some $m, n \in \mathbb{Z}$. This means $(2m, m + 3) = (2n, n + 3)$, thus $2m = 2n$. Dividing this equation by two gives $m = n$. Thus f is injective.

Next we show f is not surjective. Consider $(1, 0) \in \mathbb{Z} \times \mathbb{Z}$. If $f(n) = (1, 0)$ for some $n \in \mathbb{Z}$, then equating the first coordinates gives $2n = 1$, which has no solution $n \in \mathbb{Z}$. So no integer maps to $(1, 0)$, and f is not surjective.

6. A function $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ is defined as $f(m, n) = 3n - 4m$. Verify whether this function is injective and whether it is surjective.

This function is surjective but not injective.

First we show f is not injective. Observe that $f(0, 0) = 3(0) - 4(0) = 0$ and $f(3, 4) = 3(4) - 4(3) = 12 - 12 = 0$. Thus $f(0, 0) = f(3, 4)$ even though $(0, 0) \neq (3, 4)$, so f is not injective.

Next we show f is surjective. Take any $b \in \mathbb{Z}$. Then $(-b, -b) \in \mathbb{Z} \times \mathbb{Z}$, and $f(-b, -b) = 3(-b) - 4(-b) = -3b + 4b = b$. So every $b \in \mathbb{Z}$ equals $f(m, n)$ for some (m, n) in the domain, and f is surjective.

8. A function $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$ is defined as $f(m, n) = (m + n, 2m + n)$. Verify whether this function is injective and whether it is surjective.

This function is both injective and surjective, hence bijective.

First we show f is injective. Suppose $f(m, n) = f(k, \ell)$ for some $(m, n), (k, \ell) \in \mathbb{Z} \times \mathbb{Z}$. This means $(m + n, 2m + n) = (k + \ell, 2k + \ell)$. Equating corresponding coordinates gives the two equations $m + n = k + \ell$ and $2m + n = 2k + \ell$. Subtracting the first from the second yields $m = k$, and substituting this back into the first gives $n = \ell$. Therefore $(m, n) = (k, \ell)$, so f is injective.

Next we show f is surjective. Take any $(a, b) \in \mathbb{Z} \times \mathbb{Z}$, and let $m = b - a$ and $n = 2a - b$. So $(m, n) \in \mathbb{Z} \times \mathbb{Z}$, and $f(m, n) = ((b - a) + (2a - b), 2(b - a) + (2a - b)) = (a, b)$. So every $(a, b) \in \mathbb{Z} \times \mathbb{Z}$ equals $f(m, n)$ for some (m, n) in the domain, and f is surjective.

Being both injective and surjective, f is bijective.

10. Prove the function $f : \mathbb{R} - \{1\} \rightarrow \mathbb{R} - \{1\}$ defined by $f(x) = \left(\frac{x+1}{x-1}\right)^3$ is bijective.

Proof. First we show f is injective. Suppose $f(x) = f(y)$ for some $x, y \in \mathbb{R} - \{1\}$, that is,

$$\left(\frac{x+1}{x-1}\right)^3 = \left(\frac{y+1}{y-1}\right)^3.$$

Every real number has exactly one real cube root, so taking cube roots of both sides gives

$$\frac{x+1}{x-1} = \frac{y+1}{y-1}.$$

Since $x \neq 1$ and $y \neq 1$, both denominators are nonzero, so we may cross-multiply to get $(x+1)(y-1) = (y+1)(x-1)$. Expanding both sides,

$$xy - x + y - 1 = xy - y + x - 1.$$

Cancelling xy and -1 leaves $-x + y = -y + x$, that is, $2y = 2x$, so $x = y$. Thus f is injective.

Next we show f is surjective. Take any $b \in \mathbb{R} - \{1\}$ and let $c = \sqrt[3]{b}$. Note that $c \neq 1$, for otherwise $b = c^3 = 1$. Now let

$$x = \frac{c+1}{c-1},$$

which is defined because $c \neq 1$. Also $x \neq 1$, since $x = 1$ would give $c+1 = c-1$, which is impossible. Hence $x \in \mathbb{R} - \{1\}$. Observe that

$$x+1 = \frac{c+1}{c-1} + 1 = \frac{2c}{c-1} \quad \text{and} \quad x-1 = \frac{c+1}{c-1} - 1 = \frac{2}{c-1},$$

so $\frac{x+1}{x-1} = \frac{2c}{2} = c$, and therefore

$$f(x) = \left(\frac{x+1}{x-1}\right)^3 = c^3 = b.$$

So every $b \in \mathbb{R} - \{1\}$ equals $f(x)$ for some x in the domain, and f is surjective.

Being both injective and surjective, f is bijective. □

12. Consider the function $\theta : \{0, 1\} \times \mathbb{N} \rightarrow \mathbb{Z}$ defined as $\theta(a, b) = a - 2ab + b$. Is θ injective? Is it surjective? Bijective? Explain.

This function is bijective. Since the first coordinate a can only be 0 or 1, we compute the two cases separately: $\theta(0, b) = 0 - 2(0)b + b = b$ and $\theta(1, b) = 1 - 2b + b = 1 - b$. As b ranges over $\mathbb{N} = \{1, 2, 3, \dots\}$, the first case produces exactly the integers $1, 2, 3, \dots$, and the second produces exactly $0, -1, -2, \dots$, that is, the integers less than or equal to 0.

14. Consider the function $\theta : \mathcal{P}(\mathbb{Z}) \rightarrow \mathcal{P}(\mathbb{Z})$ defined as $\theta(X) = \overline{X}$. Is θ injective? Is it surjective? Bijective? Explain.

This function is a bijection, to show this, note that $\overline{\overline{X}} = X$ for every $X \subseteq \mathbb{Z}$.

To see that θ is injective, suppose $\theta(X) = \theta(Y)$ for some $X, Y \in \mathcal{P}(\mathbb{Z})$, that is, $\overline{X} = \overline{Y}$. Taking complements of both sides gives $\overline{\overline{X}} = \overline{\overline{Y}}$, that is, $X = Y$. Thus θ is bijective.

To see that θ is surjective, take any $Y \in \mathcal{P}(\mathbb{Z})$. Then $\overline{Y}$ is also a subset of $\mathbb{Z}$, so $\overline{Y} \in \mathcal{P}(\mathbb{Z})$, and $\theta(\overline{Y}) = \overline{\overline{Y}} = Y$.

Being both injective and surjective, θ is bijective.

16. This question concerns functions $f : \{A, B, C, D, E\} \rightarrow \{1, 2, 3, 4, 5, 6, 7\}$. How many such functions are there? How many of these functions are injective? How many are surjective? How many are bijective?

The domain has 5 elements and the codomain has 7.

A function can be determined by choosing a value in $\{1, 2, \dots, 7\}$ for each of the 5 elements of the domain, so by the multiplication principle there are $7^5 = 16807$ such functions.

Now, the 5 values must be distinct, so there are 7 choices for the image of A , then 6 for B , then 5, 4 and 3. This gives $7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 = 2520$ injective functions.

There are no surjective functions as this would require every element in the codomain to be $f(x)$ for some x in the domain, but the domain supplies only 5 values.

Because none of the functions are surjective, none of the functions are bijective.

12.3 The Pigeonhole Principle

2. Prove that if a is a natural number, then there exist two unequal natural numbers k and ℓ for which $a^k - a^\ell$ is divisible by 10.

Proof. Let $a \in \mathbb{N}$ and consider the eleven powers $a^1, a^2, a^3, \dots, a^{11}$. Dividing any integer by 10 leaves a remainder in $\{0, 1, 2, \dots, 9\}$, so there are only ten possible remainders. Because the number of powers is greater than the number of remainders, the pigeonhole principle guarantees that at least two of the powers would have the same remainder. That is, there are natural numbers k and ℓ with $1 \leq k < \ell \leq 11$ such that a^k and a^ℓ leave the same remainder when divided by 10. Writing $a^k = 10q + r$ and $a^\ell = 10q' + r$ for integers q, q' and a common remainder r , we get

$$a^k - a^\ell = (10q + r) - (10q' + r) = 10(q - q').$$

Since $q - q'$ is an integer, $10 \mid (a^k - a^\ell)$. As $k < \ell$, the numbers k and ℓ are unequal. $\square$

12.4 Composition

2. Suppose $A = \{1, 2, 3, 4\}$, $B = \{0, 1, 2\}$, $C = \{1, 2, 3\}$. Let $f : A \rightarrow B$ be

$$f = \{(1, 0), (2, 1), (3, 2), (4, 0)\},$$

and $g : B \rightarrow C$ be $g = \{(0, 1), (1, 1), (2, 3)\}$. Find $g \circ f$.

$$g \circ f = \{(1, 1), (2, 1), (3, 3), (4, 1)\}$$

4. Suppose $A = \{a, b, c\}$. Let $f : A \rightarrow A$ be the function

$$f = \{(a, c), (b, c), (c, c)\},$$

and let $g : A \rightarrow A$ be the function $g = \{(a, a), (b, b), (c, a)\}$. Find $g \circ f$ and $f \circ g$.

$$\begin{aligned} g \circ f &= \{(a, a), (b, a), (c, a)\} \\ f \circ g &= \{(a, c), (b, c), (c, c)\} \end{aligned}$$

6. Consider the functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = \frac{1}{x^2 + 1}$ and $g(x) = 3x + 2$. Find the formulas for $g \circ f$ and $f \circ g$.

$$\begin{aligned} g \circ f(x) &= \frac{3}{x^2 + 1} + 2 \\ f \circ g(x) &= \frac{1}{(3x + 2)^2 + 1} \end{aligned}$$

8. Consider the functions $f, g : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$ defined as $f(m, n) = (3m - 4n, 2m + n)$ and $g(m, n) = (5m + n, m)$. Find the formulas for $g \circ f$ and $f \circ g$.

$$\begin{aligned} g \circ f(m, n) &= g(3m - 4n, 2m + n) \\ &= (5(3m - 4n) + (2m + n), 3m - 4n) \\ &= (17m - 19n, 3m - 4n). \end{aligned}$$

$$\begin{aligned} f \circ g(m, n) &= f(5m + n, m) \\ &= (3(5m + n) - 4m, 2(5m + n) + m) \\ &= (11m + 3n, 11m + 2n). \end{aligned}$$

10. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by the formula $f(x, y) = (xy, x^3)$. Find a formula for $f \circ f$.

$$\begin{aligned} f \circ f(x, y) &= f(xy, x^3) \\ &= ((xy)(x^3), (xy)^3) \\ &= (x^4y, x^3y^3). \end{aligned}$$

12.5 Inverse Functions

2. In Exercise 9 of Section 12.2 you proved that $f : \mathbb{R} - \{2\} \rightarrow \mathbb{R} - \{5\}$ defined by $f(x) = \frac{5x+1}{x-2}$ is bijective. Now find its inverse.

Solving $y = \frac{5x+1}{x-2}$ for x gives $x = \frac{2y+1}{y-5}$, so

$$f^{-1} : \mathbb{R} - \{5\} \rightarrow \mathbb{R} - \{2\}, \quad f^{-1}(x) = \frac{2x+1}{x-5}.$$

4. The function $f : \mathbb{R} \rightarrow (0, \infty)$ defined as $f(x) = e^{x^3+1}$ is bijective. Find its inverse.

Solving $y = e^{x^3+1}$ for x gives $x = \sqrt[3]{\ln y - 1}$, so

$$f^{-1} : (0, \infty) \rightarrow \mathbb{R}, \quad f^{-1}(x) = \sqrt[3]{\ln x - 1}.$$

6. The function $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z} \times \mathbb{Z}$ defined by the formula $f(m, n) = (5m + 4n, 4m + 3n)$ is bijective. Find its inverse.

Solving $5m + 4n = a$ and $4m + 3n = b$ for m and n gives $m = -3a + 4b$ and $n = 4a - 5b$, so

$$f^{-1}(a, b) = (-3a + 4b, 4a - 5b).$$

8. Is the function $\theta : \mathcal{P}(\mathbb{Z}) \rightarrow \mathcal{P}(\mathbb{Z})$ defined as $\theta(X) = \overline{\overline{X}}$ bijective? If so, what is its inverse?

Yes. Since $\overline{\overline{X}} = X$, the function θ is its own inverse: $\theta^{-1}(X) = \overline{\overline{X}}$.

10. Consider $f : \mathbb{N} \rightarrow \mathbb{Z}$ defined as $f(n) = \frac{(-1)^n(2n-1)+1}{4}$. This function is bijective by Exercise 18 in Section 12.2. Find its inverse.

By Exercise 18 of Section 12.2, $f(2k) = k$ for $k \geq 1$ and $f(2k+1) = -k$ for $k \geq 0$. Inverting each case,

$$f^{-1} : \mathbb{Z} \rightarrow \mathbb{N}, \quad f^{-1}(c) = \begin{cases} 2c & \text{if } c \geq 1, \\ 1-2c & \text{if } c \leq 0. \end{cases}$$

12.6 Image and Preimage

2. Consider the function $f : \{1, 2, 3, 4, 5, 6, 7\} \rightarrow \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ given as

$$f = \{(1, 3), (2, 8), (3, 3), (4, 1), (5, 2), (6, 4), (7, 6)\}.$$

Find:

$$f(\{1, 2, 3\}), \quad f(\{4, 5, 6, 7\}), \quad f(\emptyset), \quad f^{-1}(\{0, 5, 9\}) \quad \text{and} \quad f^{-1}(\{0, 3, 5, 9\}).$$

4. This problem concerns functions

$$f : \{1, 2, 3, 4, 5, 6, 7, 8\} \rightarrow \{0, 1, 2, 3, 4, 5, 6\}.$$

How many such functions have the property that $|f^{-1}(\{2\})| = 4$?

6. Given a function $f : A \rightarrow B$ and a subset $Y \subseteq B$, is $f(f^{-1}(Y)) = Y$ always true? Prove or give a counterexample.

8. Given a function $f : A \rightarrow B$ and subsets $W, X \subseteq A$, then $f(W \cap X) = f(W) \cap f(X)$ is false in general. Produce a counterexample.

10. Given $f : A \rightarrow B$ and subsets $Y, Z \subseteq B$, prove $f^{-1}(Y \cap Z) = f^{-1}(Y) \cap f^{-1}(Z)$.

12. Consider $f : A \rightarrow B$. Prove that f is injective if and only if $X = f^{-1}(f(X))$ for all $X \subseteq A$. Prove that f is surjective if and only if $f(f^{-1}(Y)) = Y$ for all $Y \subseteq B$.

14. Let $f : A \rightarrow B$ be a function, and $Y \subseteq B$. Prove or disprove: $f^{-1}(f(f^{-1}(Y))) = f^{-1}(Y)$.